

SWOT Analysis

	Helpful	Harmful
Internal	Strengths 1. Strong subscriber base 2. Supportive government policies 3. Technical expertise 4. Human capital 5. Start-up incubation centres across the country 6. Faster adoption of technology	Weaknesses 1. Absence of home-grown market 2. Dependency on legacy technologies 3. Inherent lack of infrastructure 4. Lack or delayed implementation of well-defined domestic standards and policies 5. Less popular bundled telecommunication services 6. More prepaid users 7. High sales tax rates and duties for domestic manufacturers 8. Poor network coverage
External	Opportunities 1. Market for any new invention 2. Data explosion 3. Shooting number of mobile devices 4. Increased demand for better quality of services 5. Push for Indian manufacturing movement 6. Innovations that utilise commercial prospects of 4G and 5G	Threats 1. Import of low-quality devices 2. Cheap handsets by Chinese manufacturers 3. High cost for buying spectrum and developing infrastructure

"If telecom manufacturing develops in India, it would provide a huge boost to the T&M sector too."

—**Jayanth Ramachandran**, regional manager, Rohde & Schwarz India Pvt Ltd

"The telecom sector is partnering with other industry segments to build a strong, connected ecosystem."

—**Nagaraja Upadhy**, vice president of network product line, Huawei Technologies Co. Ltd

"High-speed and high-quality connectivity can be brought to rural areas by leveraging existing infrastructure."

—**Ravi Manik**, director of business development, Broadcom India and South East Asia

"Customs clearance has to improve for the volume of production to pick up."

—**Srinivasa Moorthy**, director of design engineering, D4X Technologies Pvt Ltd

"GSM is still popular in India for cost-effective voice services for business enterprises and will remain that way for the next two years at least."

—**Dhaval Bhagora**, marketing manager - telecom solutions, Matrix Comsec

"There should be a mechanism in which the telecom technology arm of the government interacts with the industry to create a product development roadmap and requirement specifications that can be taken up by product engineering companies."

—**Dhiraj Kumar**, director - systems engineering, Argus Systems

Affordable devices: A boon or a curse

A big challenge for deploying the 4G system was the lack of supporting devices. While 4G-supported phones were a luxury till last year, the second half of 2014 saw the launch of several affordable 4G handsets.

Apart from established foreign brands like Huawei, Motorola (acquired by Lenovo Group) and Xiaomi, and the previously-mentioned Indian brands, Indian market also sells cheap, low-quality, generic Chinese handsets.

In a way this is beneficial to the country as LTE rollout becomes faster with increasing demand from customers. But, Jayanth Ramachandran, regional manager, Rohde & Schwarz India Pvt Ltd, points out that increased usage of low-quality devices would load the network, leading to reduced speed and performance. How do we solve this problem?

Need for customised standards

Increased import of low-quality devices is due to the fact that there

are no stringent rules and regulations regarding the minimum quality requirement to sell a product in the Indian market.

In addition to this, import duties here are low. This negatively affects the performance of the network.

Key performance indicators for a device need to be set higher. Nagaraja Upadhy, vice president of network product line, Huawei Technologies Co. Ltd, India, however, has a different opinion. He stresses that the need of the hour is development of a 4G ecosystem in India, which is possible only if more people can access 4G technology at a lower cost. He also mentions that affordable handsets need not always be of low quality.

Effect on electronics industry

The effect on some segments of electronics industry by advancements in telecommunication technologies are described below.

PCB industry. The advanced telecom technologies lead to increased feature sets with lowest form factors, resulting in complex designs. Due to this, printed circuit board (PCB) designers and manufacturers specialising in this field require adoption of high-performance PCB design and circuit simulation tools, additional design analyses, manufacturing verification tools and an upgradation of existing design methodologies.

Another challenge is electronic systems' reliability since it is very important to telecom and data communication service providers.

However, reliability of a PCB is dependent on its thermal performance, which increases with complexity and reduced computing time. This makes sophisticated system-level thermal design and management a vital part of designing telecom equipment.

"A PCB designer has to manage power dissipation and optimise the thermal design by modelling and analysing PCB temperatures using design analysis tools," says Dhiraj Kumar, director - systems engineering, Argus Systems.